Bacterial Whole Genome Sequencing Analysis Report

Patient name	**:**	**XXX**	**PIN**	**:**	**XXX**
Sample source	**:**	**XXX**	**Sample number**	**:**	**XXX**
Referring Clinician	**:**	**XXX**	**Sample collection date**	**:**	**XXX**
Specimen	**:**	**Culture plate**	**Sample received date**	**:**	**XXX**
			Report date	**:**	**XXX**

Methodology

Read quality check

Initially, we checked the following parameters from the sample Fastq file - Base quality score distribution, sequence quality score distribution, average base content per read, GC distribution in the reads, PCR amplification issue, overrepresented sequences, and adapters.

Based on the quality report, the Fastq files were trimmed to retain high-quality sequences and the low-quality sequence reads were excluded from the analysis. The adapter trimming was performed using the Fastq-mcf tool (version- 1.04.803).

De novo assembly

The quality filtered and adapter trimmed reads were used for Denovo assembly using Spades, to get the primarily assembled genome.

ORF prediction and annotation

Then Primarily assembled was used for ORF prediction and ORF annotation using Prodigal.

Genome annotation and functional annotation

The Primarily assembled genome was annotated using prokka to predict genes, CDS etc. And the functional annotation was performed using Emapper.

Blast

The NCBI blast was performed for the primarily assembled genome to predict the closest reference genome of each sample.

Reference-guided assembly

The genome fasta of the predominant species (predicted from above step) was downloaded from NCBI database and reference-guided assembly was performed by aligning human unaligned reads and consensus genome fasta was generated using samtools and Bcftools.

Bioinformatics Analysis Results

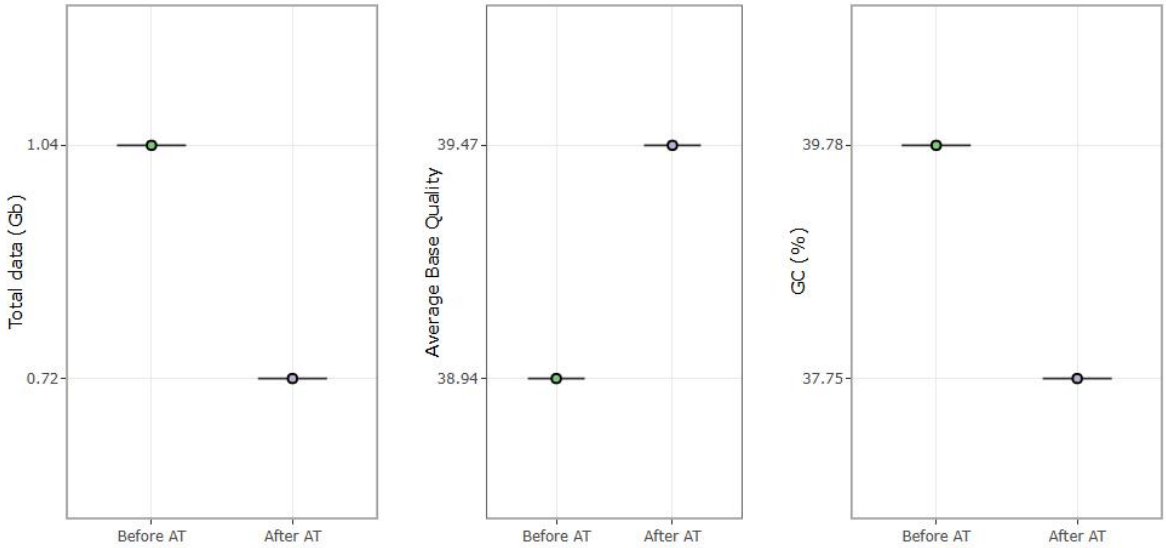
Raw Data Summary

Samples were taken for whole genome analysis and around 1.04 Gb data was generated for the sample. Q30% was above 94.28% and average GC% is around 39.78%. Kindly refer to the QC report for further details. The quality check for the sample after adapter trimming has been performed and the adapter trimmed sequences are taken for downstream analysis.

Table 1: Data summary

Sample name	Total reads (R1+R2)	GC %	Total Data >= Q30(%)	Data Size (in GB)	Average Read Length	Average Base Quality (phred)
XXX	6,858,418	39.78	94.28	1.04(Gb)	151	38.94

Figure 1: plot showing total data, average base quality and GC % for all the samples before and after adapter trimming.



De novo assembly

De novo assembly was carried out using spades assembler (v3.11.1). Assembly is performed with k-mer size 55 using de-Bruijn graph method. The assembled scaffolds were taken for further downstream analysis. The distribution of scaffolds, N50 values are given in the Table 2.

Table 2: De novo assembly stats

Sample name	K-mer	No of scaffolds(>=0_bp)	N50	N90	L50	L90	Largest contig
XXX	55	167	54360	17376	25	86	368,675

Blast

The NCBI blast was performed for the primarily assembled genome to predict the closest reference genome of each sample.

For complete blast result please refer the supplementary

Table 3: Blast Predominant species summary based on length and No. of contigs

Sample Name	Species
XXX	*CP070490.1 Lysinibacillus fusiformis strain NEB1292 chromosome, complete genome*

Reference guided Assembly

The adapter trimmed reads were aligned to *Lysinibacillus fusiformis strain NEB1292 chromosome,* reference *genome*. The aligned consensus fasta were generated using the samtools tools .

AMR Gene Prediction

AMR prediction performed using the RGI (Resistance Gene Identifier) tool. The application uses reference data from the Comprehensive Antibiotic Resistance Database (CARD), it is a biological database that collects and organizes reference information on antimicrobial resistance genes, proteins and phenotypes.

Please refer the supplementary AMR results for detailed information.

Genome Annotation

Genome annotation was done using Prokka (1.14.6). The assembled genome was taken for genome annotation.

Table 4: Genome Annotation Table

Sample ID	CDS	rRNA	tRNA	tmRNA
XXX	4866	34	108	1

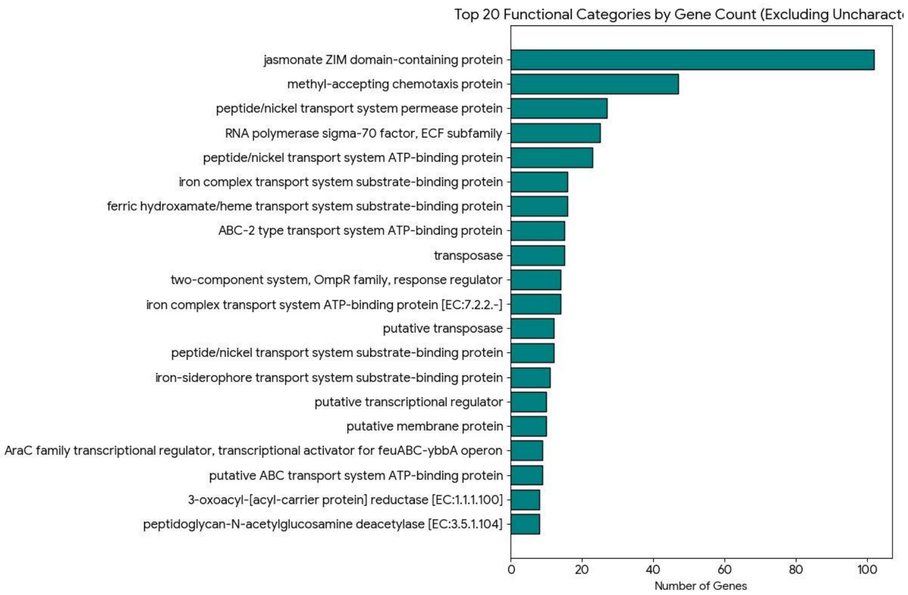
Variants Calling and Annotation

Variant calling was done for the samples using GATK-haplotypecaller and further annotated using SnpEff. The list of variants and gene annotation is provided in the supplementary files. Please refer to Supplementary files.

Pathways & Annotation

We have provided KEGG Functional annotation results as supplementary files.

Figure 2: The top20 Functional Categories by gene count



Supplementary Files:

Supplemenatry1.1: ORF Sequences (fasta file)
Supplemenatry1.2: Blast results (xls)
Supplemenatry1.3: Denovo assembled fasta (fasta files)
Supplemenatry1.4: Consensus fasta (fasta files)
Supplemenatry1.5: VCF file
Supplemenatry1.6: Functional_Annotation (xls)
Supplemenatry1.7: Genome_Annotation
Supplemenatry1.8: AMR results

References:

Li H. and Durbin R. (2009) Fast and accurate short read alignment with Burrows-Wheeler Transform. Bioinformatics, 25:1754-60. [PMID: 19451168]

Nurk S, Meleshko D, Korobeynikov A, Pevzner PA. metaSPAdes: a new versatile metagenomic assembler. *Genome Res*. 2017;27(5):824-834.

Hyatt D, Chen GL, Locascio PF, Land ML, Larimer FW, Hauser LJ. Prodigal: prokaryotic gene recognition and translation initiation site identification. *BMC Bioinformatics*. 2010;11:119. Published 2010 Mar 8. doi:10.1186/1471-2105-11-119

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